yss1的解题思路

题目信息

题目名	类型	难度
yss1	DS	入门

FLAG

- 动态flag

知识点

1. 数据识别与分类

解题步骤

1、观察附件提供的data.txt发现是一堆混乱的数据，然后查看数据格式规范，依据数据格式规范编写poc.py，实现正确提取身份证号码和mac地址，poc如下。

```
import re
import datetime

# 身份证号校验函数
def is_valid_idcard(idcard):
    # 检查长度
    if len(idcard) != 18:
        return False

    # 检查出生日期是否合法
    try:
        year = int(idcard[6:10])
        month = int(idcard[10:12])
        day = int(idcard[12:14])

        # 检查年份是否在合理范围内(1900-2023)
        if not (1900 <= year <= datetime.datetime.now().year):
            return False

        # 验证日期是否有效
        datetime.datetime(year, month, day)
    except ValueError:
        return False

    # 验证校验码
    factors = [7, 9, 10, 5, 8, 4, 2, 1, 6, 3, 7, 9, 10, 5, 8, 4, 2]
    checksum_chars = ['1', '0', 'X', '9', '8', '7', '6', '5', '4', '3', '2']

    checksum = 0
    for i in range(17):
```

```python
        checksum += int(idcard[i]) * factors[i]

    correct_checksum = checksum_chars[checksum % 11]
    return idcard[17].upper() == correct_checksum

# 正则表达式
regex_patterns = {
    # 身份证号：18位数字，最后一位可能是X
    "idcard": re.compile(r"\b(\d{17}[\dXx])\b"),
    # MAC地址：必须以DE:AD:BE开头的MAC地址
    "mac": re.compile(r"\b(DE:AD:BE:[0-9A-Fa-f]{2}:[0-9A-Fa-f]{2}:[0-9A-Fa-f]{2})\b"),
}

# 主提取函数
def extract_sensitive_data(filepath):
    results = set()
    with open(filepath, 'r', encoding='utf-8') as f:
        for line in f:
            # 匹配每种类型的数据
            for category, pattern in regex_patterns.items():
                matches = pattern.findall(line)
                for match in matches:
                    # 由于正则表达式使用了捕获组，match可能是元组
                    if isinstance(match, tuple):
                        value = match[0]  # 使用第一个捕获组
                    else:
                        value = match

                    # 验证身份证合法性
                    if category == "idcard" and not is_valid_idcard(value):
                        continue

                    results.add(f"{category}_{value}")

    return sorted(results)

# 写入提取结果
def write_results(results, output_path="answer.txt"):
    with open(output_path, "w", encoding="utf-8") as f:
        for line in results:
            f.write(line + "\n")
    print(f"□ 提取完成，结果已保存到：{output_path}")

if __name__ == "__main__":
    input_file = "data.txt"
    output_file = "answer.txt"
    extracted = extract_sensitive_data(input_file)
    write_results(extracted, output_file)
```

2、运行poc.py，查看跑出来的answer.txt，结果如下图所示。

```
idcard_110004201907088197
idcard_110113202212233243
idcard_110145197712024760
idcard_110347196710041072
idcard_11035619450424540X
idcard_110377198408082620
idcard_110417194103135702
idcard_110426200704098470
idcard_110633195504079161
idcard_110649195101182819
idcard_110757201811087780
idcard_110770195404242563
idcard_110809202012222922
idcard_11091219630702286X
idcard_110944196611034559
idcard_111045198207116496
idcard_111143198105087818
idcard_111205194004304229
idcard_111231200808298179
idcard_111237198409146264
idcard_111297201801156396
idcard_11145619860701473X
idcard_111462197002087381
idcard_111553198504128204
idcard_111567202509284545
idcard_111576194211307854
idcard_111582194709131945
idcard_111583194902244665
idcard_111584199804053856
idcard_111653202306274213
```

3、最后当问验证靶机，提交提取的answer.txt,发现成功通过并拿到flag。